

Deadline: Nov. 21th, 2022

1. Why the conductivity of Metal increase with decrease in temperature, semiconductor increase with increase in temperature?
2. Why gallium arsenide is used to make optical devices such as LEDs, whereas silicon is not?
3. Silicon wafer is made from what kind of raw material?
4. What is the advantage of using a high dielectric-constant material as the gate oxide of a MOSFET.
5. Draw a MOSFET transistor, label gate, source, and drain regions. Explain how MOSFET works.
6. Please read the textbook carefully and give a brief description for each step of the CMOS IC fabrication as illustrated in the following figure.

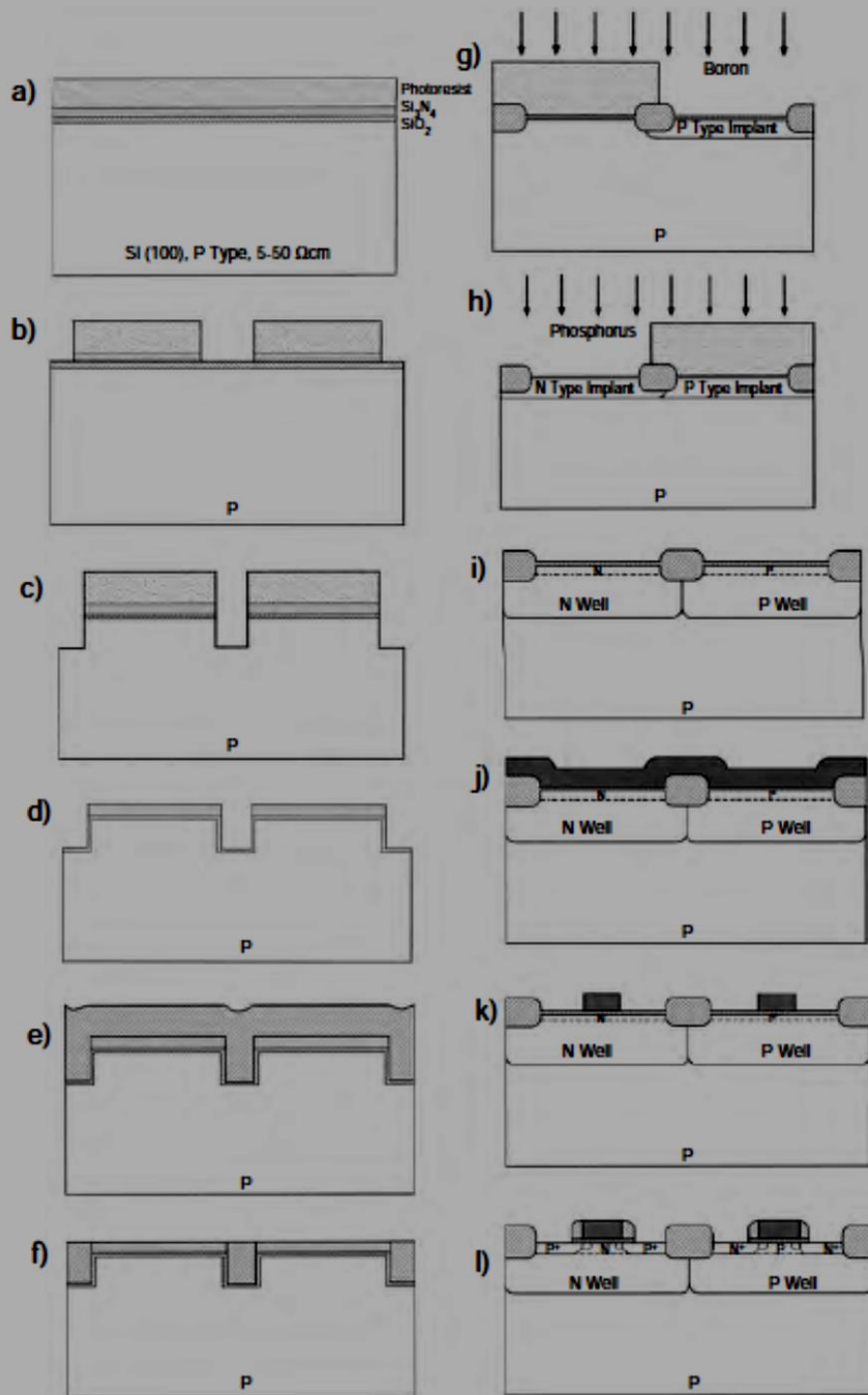


Figure 4.25. The steps involved in fabricating a CMOS IC. Reproduced with permission from Plummer, J. D.; Deal, M. D.; Griffin, P. B. *Silicon VLSI Technology*, Prentice-Hall: New York, 2000.

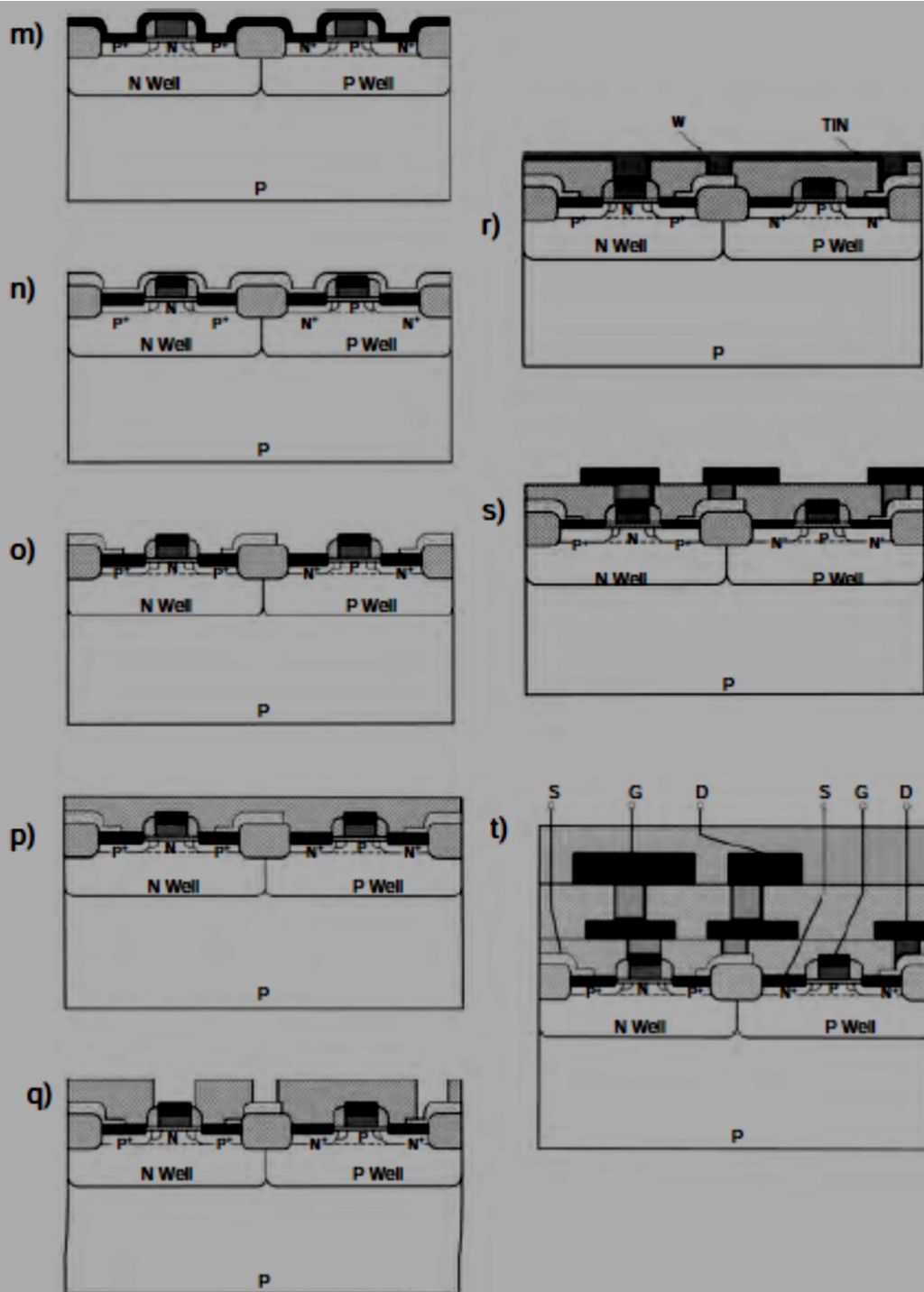


Figure 4.25. Continued